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The evolution of Game Software Engineering: A Systematic Literature Review

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Take pride in what is sure to die.

Abstract

Context: Game Software Engineering (GSE) is a branch of Software Engineering (SE) specialised in the development of videogames. Its academic relevance has been steadily growing for years, making it an independent scientific domain.

Objective: This study intends to update a previous one to evaluate the current state of the art in software engineering, with a focus on industry-scale computer games.

Method: A Systematic Literature Review (SLR) was performed to find and analyse relevant studies. A total of 40 papers were included and used to extract all the relevant information.

Results: The amount of research activities conducted in the GSE field is ever increasing, producing a bit more than half the papers found by the previous study in a time frame a third of the size. The general quality of the studies has increased, but authors still need to be more rigorous on critically examining their own role. Testing is the most investigated research topic, with a focus on automating processes, followed by system structures.

Conclusion: This study shows that GSE continues to mature as an independent research field characterized by growing empirical research and stronger industry engagement. The results confirm a divergence between GSE and traditional SE and reveal limitations in the existing classification methods. Overall, GSE emerges as a fertile domain for advancing software engineering knowledge and practice.

Keywords: Computer games, Video games, Game software engineering, Systematic literature review, SLR, Industry-scale, Review update

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CAPITOLO 1

Introduction

Videogames have evolved dramatically over the past decades, becoming one of the most complex and resource intensive software products to develop. Modern titles integrate a unique blend of programming, animation, audio engineering, artificial intelligence, physics simulation, and user experience considerations. All of these disciplines, combined with the pressure of tight schedules and deadlines, place extraordinary demands on development teams. Studios must coordinate large, multidisciplinary groups where people with various backgrounds and attitudes must work together, often iterating at a rapid pace to refine gameplay.

Within this context, Game Software Engineering (GSE) emerged as a specialised branch of Software Engineering (SE) to address challenges that traditional SE practices could not fully capture. Over time, the gap between SE and GSE has widened, as videogame development increasingly relies on fast prototyping, creative iterations, new testing approaches, and player experience. GSE research now encompasses topics like automated playtesting, performance optimisation under real-time constraints, game engine customisation, and creative iteration cycles; areas where traditional SE researchers do not focus.

This study is a Systematic Literature Review (SLR) that updates the analysis conducted by Chueca et al. [1], therefore, it follows their methodology while introducing

some adjustments. As the industry grows and academic interest intensifies, GSE has matured into a coherent research domain with its own methods, priorities, and empirical evidence. Several studies argue that GSE should be recognised as distinct from SE [1, 2, 3, 4, 5], and the expanding body of literature reinforces this position. This work, too, intends to contribute to this discussion and support this claim by providing an updated and comprehensive overview of the field.

1.1 State of the industry

The revenue of the gaming industry increases every year and is expected to reach \$197 billion in 2025, growing +7.5% year-over-year and exceeding previous expectations; projections also suggest that this upward trend will continue in the coming years [6]. The global player base is currently estimated at 3.6 billion and is expected to reach almost 3.9 billion by 2028 [7]. This sustained growth has positioned videogames as one of the largest entertainment sectors worldwide, surpassing films and music combined.

However, all that glitters is not gold. Since 2022, the general landscape of the gaming industry has shifted due to widespread layoffs: the numbers increased in 2023, peaked in 2024, and only began to ease in 2025 [8]. These cuts affected companies of all sizes, from major publishers to independent studios, and led to delayed or cancelled projects, studio closures, and thousands of employees losing their jobs [9]. Instability has been attributed to multiple factors, including increased production costs, market saturation, and financial risks associated with AAA development.

At the same time, the industry is experiencing a technological transformation, with new advancements in virtual and extended reality, cloud gaming, and AI-assisted content creation. This situation makes the field of GSE even more relevant, as there is a strong and evident need for robust domain-specific software engineering practices.

1.2 Goals and contributions

The main goal of this paper is to clarify the current state of research on software engineering techniques, approaches, and methods applied to computer games, and to compare these findings with the previous work conducted by Chueca et al. [1]. By updating and extending their analysis, this study will provide a more comprehensive and up-to-date picture of how GSE has evolved in recent years. This overview will support future research by identifying active areas of investigation, highlighting methodological trends, and revealing gaps that remain under-explored.

A second goal is to assess whether the volume, diversity, and maturity of current GSE research indicate that the field is increasingly distinct from traditional SE. Understanding this distinction is important, as it can shape future research agendas and industry practices. The findings of this SLR will therefore help determine whether GSE should be considered an independent domain with its own identity and challenges.

Lastly, this work also contributes to the narrowing of the gap between industry and academia. A consolidated view of the literature can guide practitioners towards problems that matter in real production environments. The dataset produced through this review contains high quality studies that show the richness and potential of the GSE domain, offering a solid foundation for new investigations, tools, and empirical work.

1.3 Structure of this work

The study is structured as follows: Section 2 provides some information on systematic literature reviews, their presence in the field of GSE, and the reasons why this work was carried out. Section 3 addresses the research method, explaining in detail each research question, the search strategy used, and the data extraction process. Section 4 presents the results obtained from the data extracted from the included studies. Section 5 presents a detailed analysis of the results. Section 6 presents open problems and possible future directions for this field. Section 7 presents possible

threats to the validity of this review. Finally, Section 8 concludes the study report and identifies the next steps for future work.

To ensure transparency and reproducibility, all the data gathered during the steps of this review are publicly available on this GitHub repository¹.

¹<https://github.com/PDant17/Thesis-OnlineAppendix>

CAPITOLO 2

Background

As stated by Ralph and Baltes [10], systematic reviews are "[a] category of secondary research in which primary studies are selected according to a systematic, replicable process". Many aspects of their conclusions can be extrapolated and adapted from traditional SE to GSE. In fact, both fields need studies that can synthesise the great amount of research produced over the years and give concrete recommendations to practitioners.

2.1 State of secondary research

The lack of existing SLRs was previously discussed by Chueca et al. [1], who retrieved 22 studies that deal with videogames as software from 2010 to 2021: only one of them addressed software and its engineering applied to videogames, instead of focusing on a specific topic of GSE, and was written by Ampatzoglou and Stamelos in 2010 [3]. Literature reviews like these are still scarce to this day.

In recent years, various works dealing with GSE have been published, but analysed particular aspects such as education (Landáburu et al. [11]), gamification (Ristiano et al. [12]) or serious games (Damianova and Berrezueta-Guzman [13]). It

seems only natural to continue the work previously done by Chueca et al. [1] and evaluate the current state of the art in software engineering.

2.2 Assessing the need for an update

Mendes et al. proposed a framework to determine whether to update or not an existing SLR in SE [14]. The process is divided into three steps: the first one addresses how current the SLR is, the second one focuses on identifying if new methods are proposed and/or new studies got published, and the last one focuses on whether their adoption/inclusion would change the conclusions or the credibility of the previous study.

Since the GSE field evolves rapidly, determining the state of the art in Software Engineering for industry-scale computer games is a relevant matter. The work of Chueca et al. [1] had good visibility, since it received 38 citations in 3 years, with an average of 12.66 citations per year; this is well above the proposed threshold of 6.82 citations per year. The number of published studies is growing year by year and many of them may contain information on new research trends. There is a high chance that their analysis might result in the discovery of unforeseen directions emerging in this field. Even if this was not the case, providing the latest data available on the state of the art is still a valid reason, given how fast-paced the gaming industry is.

CAPITOLO 3

Review method

Since this study aims to update an existing Systematic Literature Review (SLR), the structure of the work and the steps taken will be very similar to those of Chueca et al. [1]. The focus on industry-scale computer games is maintained, which means that games and tools developed in industrial practice or for commercial purposes will be considered, while academic prototypes will not.

3.1 Research Questions

The research questions will remain the same as before to make it easier for comparisons and will be the following five:

Q RQ₁. *Has the intensity of the research activity on software engineering for industry-scale computer game development changed over the years?*

Q RQ₂. *How have software engineering research topics in the domain of industry-scale computer games changed over the years?*

Q RQ₃. *How have research approaches in the domain of industry-scale computer games changed over the years?*

Q RQ₄. *How have empirical research methods in the domain of industry-scale computer games changed over the years?*

Q RQ₅. *What is the quality of industry-scale computer game research currently?*

RQ1 has the purpose of showing whether there is an increase or a decrease in research volume over the years. This will be done by direct comparison between the retrieved research from 2009 to 2021 and from 2022 to 2025.

RQ2, RQ3, and RQ4 will be answered by analysing the data extracted from the selected studies. The evidence obtained will then be compared to the data obtained previously by Chueca et al. [1].

The system used to classify the research topic for each study is the 2012 ACM Computing Classification System (CCS), which is the de facto standard classification system for the computing field [15].

The classification of the research approaches for each study was done according to the classification proposed by Glass et al. [16]. Descriptive research typically describes systems, methods, or tools, and an example would be an SLR. Exploratory research aims to discover or understand a phenomenon not yet well-defined and determine the best data collection method and research design. Empirical research produces its findings through direct or indirect observation of real subjects.

Empirical studies can be further classified in relation to the method used into three categories, according to Wohlin et al. [17]. Experiments are set in a highly controlled environment and have subjects performing some tasks, so they are appropriate for exploring relationships between variables. Surveys ask subjects to complete questionnaires and are typically used if the method under study has been in use for a while. Case studies monitor one (or multiple) project, activity, or assignment; they are similar to experiments but use uncontrolled or semi-structured settings.

Lastly, RQ5 is used to measure the quality of the studies and highlight the presence of any common problem in the current research.

3.2 Literature search strategy

The search strategy used has some slight differences from the previous work by Chueca et al. [1]. The Databases considered for the survey are ACM Digital Library, Scopus, and IEEE. Two databases like ACM and IEEE, in conjunction with a single indexing database, Scopus, should be enough to ensure full coverage of the main publication venues. As such, Web of Science was not included in this list. Regarding the search strategy itself, the search string remained the same. It was constructed using the PICO (Population, Intervention, Comparison, Outcome) structure proposed by Kitchenham and Charters [18]:

- **Population** refers to the application area. In this case its computer games, but the synonym videogame was considered too. Variations such as singular/plural and the alternative spelling "video game" were considered.
- **Intervention** refers to procedures, methodologies, technologies, and tools used to create software. In this case, it corresponds to 'engineering', 'requirements', 'coding', 'testing', etc. The word 'software' was added before each term, as they would otherwise be too broad and refer to concepts outside the SE field.
- **Comparison** refers to the procedures, methodologies, technologies, or tools that are compared with the intervention; thus, it is not applicable in this case.
- **Outcome** refers to factors that should be important to practitioners. In this case, this was done manually because the word 'industrial' may or may not appear in relevant studies. The same applies to this update.

The Boolean operator OR combined terms inside each criterion, while the Boolean operator AND connected Population and Intervention. The terms of the former must appear in either the Title, the Abstract, or the Keywords of the study, while the terms of the latter can be anywhere. The search string is the following, but was slightly adjusted to suit the grammar of each database:

TITLE-ABS-KEY ("computer game" OR "computer games" OR "video game" OR "video games" OR "videogame" OR "videogames") AND ALL ("software engineering" OR "software development" OR "software requirements" OR "software design" OR "software coding" OR "software testing" OR "software verification" OR "software evolution" OR "software maintenance")

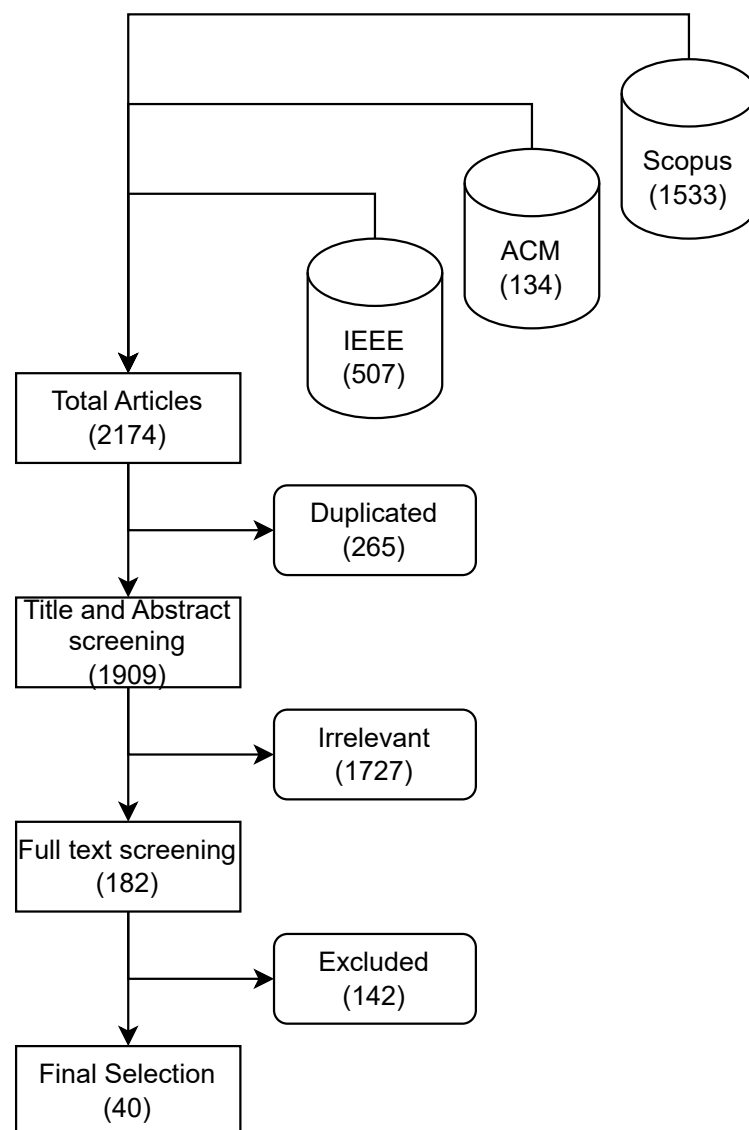
Figure 3.1 shows the steps taken to select the relevant studies: it is worth noting that the Covidence¹ tool was used to support this whole phase.

1. The search string was used on the chosen databases and the results were loaded on Covidence.
2. Covidence automatically detected 265 studies as duplicates, so the total number of studies decreased from 2174 to 1909.
3. The first screening phase was conducted by applying inclusion/exclusion criteria to the titles, abstracts, and keywords of each study leaving 182 studies left.
4. The same criteria were then applied to the full text of each study, leaving a final selection of 40 relevant studies.
5. As their final step, Chueca et al. [1] performed backward snowballing to find possible missing studies, finding 8 more. This step was not performed in this work, due to time constraints.

The selection criteria were used to evaluate the relevance of each study: they were discarded if any exclusion criteria were met and were included only if all inclusion criteria were met. These are the exclusion criteria used:

- Studies that were duplicates of other studies.
- Studies that were not written in English.
- Studies that were not published online between 2022 and 2025.

¹<https://www.covidence.org/>

**Figure 3.1:** Overview of the search process

- Studies presenting non-peer-reviewed material.
- Studies presenting peer-reviewed but not published in journals, conferences, or workshops.
- Studies that were summaries of conferences/editorials.
- Non-primary studies.

The only difference between this work and the one previously done by Chueca et al. [1] is the time window: since they covered the years from 2009 to 2021, both ends fully included, the starting point for this update had to be 2022. The inclusion criteria used, on the other hand, are exactly the same:

- Studies that did not focus on the social and educational impact of videogames, such as serious games.
- Studies that did not focus on Artificial Intelligence (AI).
- Studies that did not focus on Content Creation.
- Studies in the field of Software Engineering.
- Studies that focused on software engineering applied to industry-scale computer games development.

Cheng et al. [19] explained that the term serious game "has become a popular and particular term referring to any kind of video game-based learning and training (e.g., business, military, medical, marketing) or the so-called edutainment". They generally focus on the health or social implications of games, rather than the software engineering behind them.

Studies that dealt with AI and Content Creation were also excluded, as they generally focus on the applications and the content itself rather than the methodologies used to develop them.

Since only primary studies were considered, posters and summaries were also excluded.

The final selection consists of 40 studies that will be presented in Section 3.3.

N	Question
Q1	Is there a rationale provided for why the study was undertaken?
Q2	Is there an adequate description of the context (e.g., industry, laboratory setting, products used, etc.) in which the research was carried out?
Q3	Is there a justification and description for the research design?
Q4	Is there a clear statement of the findings, including data that supports findings?
Q5	Did the researcher(s) critically examine their own role, potential bias, and influence during the study?
Q6	Are limitations and credibility of the study discussed explicitly?

Tabella 3.1: Questions to assess study quality.

3.3 Quality assessment

Similarly to what Chueca et al. [1] did, the quality assessment of the included studies does not refer to the actual quality of the research itself, but to the quality of the reporting. The quality assessment questions will remain the same and are reported in Table 3.1. To avoid neglecting statements where the authors give limited information, a three-point scale is used, similarly to that of Ali et al. [20]. Each answer will have a correspondent numerical value ('Yes' means 1, 'To Some Extent' means 0.5, and 'No' means 0), and the final score of a study will be the sum of the scores received for each question.

3.4 Data extraction

Table 3.2 shows the data collected from each study and how they relate to the research questions. F1, F3, F6 and F7 are necessary information for every paper. F2 helps with every research question, since knowing the publication year is mandatory to make comparisons and talk about trends. F4 and F5 are needed to highlight the most active research publishers. F8 records the quality score for each study. F9 records the research topic according to the 2012 ACM Computing Classification System (CCS).

N	Field	Research question
F1	Author(s)	n/a
F2	Year	RQ1, RQ2, RQ3, RQ4, RQ5
F3	Title	n/a
F4	Venue	RQ1
F5	Publisher	RQ1
F6	Keywords	n/a
F7	Abstract	n/a
F8	Quality score	RQ5
F9	SE research topic	RQ2
F10	Research approach	RQ3
F11	Empirical research method	RQ4
F12	Limitations	RQ5

Tabella 3.2: Data collection form.

F10 and F11 record the approach taken for each study. Finally, F12 records limitations or threats addressed by the authors of each study.

This section also clarifies the classification of each study concerning the software engineering issue it involves. As mentioned in Section 3.1, the only system used is the 2012 ACM CCS. Chueca et al. [1] used it, along with the older 1998 version, to simplify the comparison process for future works and to understand how appropriate it is for GSE. The system has a tree structure, so the list of topics shown in Table 4.1 is not complete, as each of them includes more subtopics. The chosen degree of depth ensures that the groups made are representative enough. The topics were assigned during the full text review phase and it must be noted that if a study was already classified, then that topic was maintained.

Results

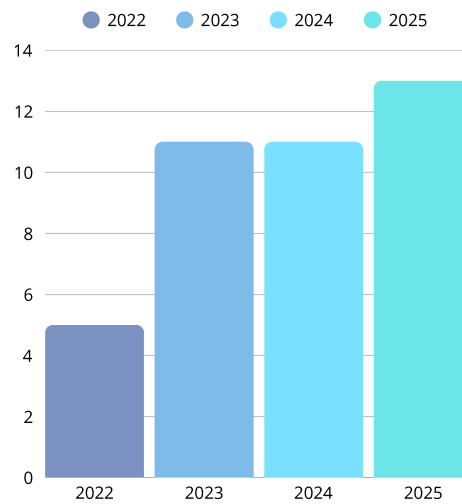
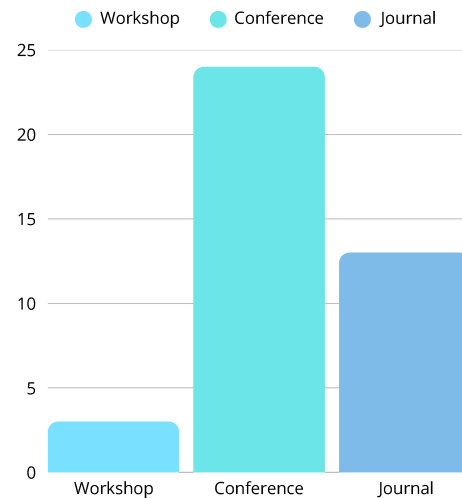
This section presents the data extracted from the 40 studies included in the Literature Review and compares them directly with the results of Chueca et al [1], since the criteria used are the same.

4.1 RQ1: Research intensity

Q RQ₁. *Has the intensity of the research activity on software engineering for industry-scale computer game development changed over the years?*

This can be directly answered by looking at the number of papers included in this review and confronting them with those from 2009 to 2021. This dataset counts 40 studies in a time span of 4 years, from 2022 to 2025, as opposed to the 98 included from 2009 to 2021, a time span of 13 years. This means that the average number of industry-scale studies per year increased from 7.53 to 10.

It may also be worth considering the total number of unique studies retrieved after querying the databases to provide information on the general volume of research. Chueca et al. [1] retrieved 2974 unique studies from 2009 to 2021, for an average of 228.76 studies per year, while the same query, from 2022 to 2025 and without using Web of Science, retrieved 1909 unique studies, for an average of 477.25 studies per

**Figura 4.1:** Research per year**Figura 4.2:** Publication types

year. This substantial increase in yearly publications demonstrates that the broader SE research landscape is highly active and that the field is expanding both in scope and scholarly engagement.

Figure 4.1 shows the number of articles per year, confirming the rising trend, while Figure 4.2 shows the types of publication: Journal (13), Conference (24) or Workshop (3). Furthermore, Figure 4.3 shows the number of unique publishers found.

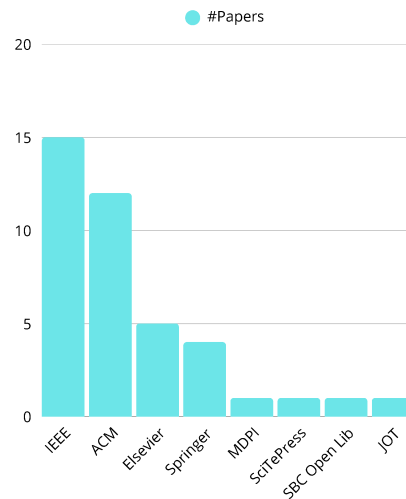


Figura 4.3: Research per publisher

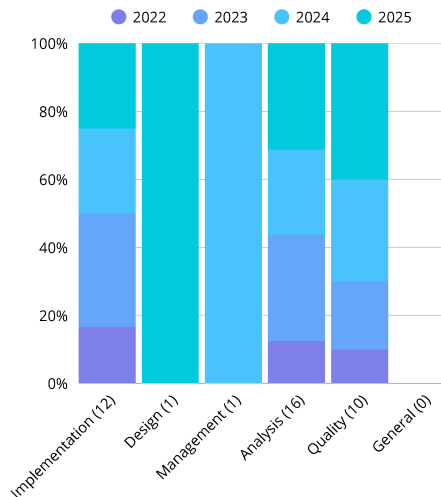


Figura 4.4: Research topics growth timeline.

4.2 RQ2: Research topics

Q RQ₂. *How have software engineering research topics in the domain of industry-scale computer games changed over the years?*

Figure 4.4 is a chart showing the distribution of generic topics over time, with 3 of them picked almost equally and the other 3 almost absent. The method of assigning generic topics was defined by Ampatzoglou and Stamelos [3] and was based on the topics of the 1998 ACM CCS.

The classification according to the 2012 ACM CCS shows unevenly distributed studies with one topic in particular dominating over all the others. Table 4.1 shows

the data on the classification of topics from both this study and the previous one conducted by Chueca et al [1], using absolute values and percentages to allow better comparisons.

The next part of this section is dedicated to a description of each study included in the review. According to Budgen et al [21], the primary studies of a mapping study should be associated with the research topic they address, so descriptions will be grouped based on the 2012 ACM CCS. To give more clarity, the studies included in the review are part of a separate bibliography and present an 'S' as suffix.

4.2.1 Contextual software domains

In [S1], the authors propose a comprehensive method to measure the power consumption of videogames.

4.2.2 Software system structures

In 2 studies [S2, S3], the authors propose new architectures to improve Quality of Experience (QoE) for players.

In 3 studies [S4, S5, S6], the authors analyse the structure of game engines, showing the relationships between their subsystems.

In 1 study [S7], the authors test whether bug localisation is easier in Model-Driven Development (MDD) or Code-centric Development (CcD).

In 1 study [S8], the authors use graph properties to measure the *comprehensive difficulty* of enemies.

4.2.3 Extra-functional properties

In 1 study [S9], the authors investigate awareness, knowledge, and priorities regarding security in the videogame development industry.

4.2.4 General programming languages

In 1 study [S10], the authors evaluated the differences in energy consumption between visual scripting models and C++.

Software Engineering Topic	2022-2025		2009-2021	
5.1.1 Contextual software domains	1	2.50%	4	4.04%
5.1.2 Software system structures	7	17.50%	14	14.14%
5.1.3 Software functional properties	0	0.00%	2	2.02%
5.1.4 Extra-functional properties	1	2.50%	3	3.03%
5.2.1 General programming languages	2	5.00%	0	0.00%
5.2.2 Formal language definitions	0	0.00%	1	1.01%
5.2.3 Compilers	0	0.00%	0	0.00%
5.2.4 Context specific languages	0	0.00%	3	3.03%
5.2.5 System description languages	0	0.00%	2	2.02%
5.2.6 Development frameworks and environments	0	0.00%	3	3.03%
5.2.7 Software configuration management and version control systems	0	0.00%	0	0.00%
5.2.8 Software libraries and repositories	0	0.00%	0	0.00%
5.2.9 Software maintenance tools	0	0.00%	0	0.00%
5.3.1 Designing software	1	2.50%	14	14.14%
5.3.2 Software development process management	2	5.00%	13	13.27%
5.3.3 Software development techniques	3	7.50%	9	9.09%
5.3.4 Software verification and validation	21	52.50%	26	26.26%
5.3.5 Software post-development issues	2	5.00%	4	4.04%
5.3.6 Collaboration in software development	0	0.00%	0	0.00%
5.3.7 Search-based software engineering	0	0.00%	0	0.00%

Tabella 4.1: GSE research topics CCS 2012.

In 1 study [S11], the authors provide a dataset of parsed Blueprint graphs extracted from Unreal Engine.

4.2.5 Designing software

In 1 study [S12], the authors define a catalogue of bad smells in videogame development and assess its relevance through a survey with professionals.

4.2.6 Software development process management

In 1 study [S13], the authors analyse postmortems to map project management anti-patterns to anti-patterns reported by game developers in the grey literature.

In 1 study [S14], the authors introduce the concept of Behaviour-Driven Development (BDD) and show various ways of applying it to videogame development.

4.2.7 Software development techniques

In 2 studies [S15, S16], the authors evaluate the benefits of Software Product Lines (SPL) and compare their use with Clone and Own (CaO).

In 1 study [S17], the authors analyse reuse practices in videogame development, highlighting both their significance and limitations.

4.2.8 Software verification and validation

In 1 study [S18], the authors present a new tool responsible for test selection and defect prevention.

In 1 study [S19], the authors present a tool specifically designed to detect bugs related to camera clipping.

In 2 studies [S20, S21], the authors integrate Behaviour-Driven Development (BDD) into videogame development to automate various aspects of testing.

In 1 study [S22], the authors implement a new algorithm that automates reachability testing.

In 1 study [S23], the authors extend the capabilities of an existing tool with a plugin that simplifies scriptless testing.

In 1 study [S24], the authors analyse the performance of a new method to predict build outcomes.

In 1 study [S25], the authors analyse the impact that changes in products have on artefacts made by other teams using dependency graphs.

In 2 studies [S26, S27], the authors describe 2 tools that help extract text from game screenshots and spoken audio and run automated tests to check for localisation issues. In a similar study [S28], the authors propose a new tool that efficiently detects multilingual text glitches and generates reports of the findings.

In 1 study [S29], the authors propose a method to map requirements to code by reproducing Non-Playable Characters (NPCs) actions.

In 1 study [S30], the authors present a new technique for regression testing selection that uses State Transition Graphs (STG).

In 2 studies [S31, S32], the authors deal with visual programming: in the first one, they build new models that encompass visual code metrics and compare them with traditional ones; in the second one, they present a modified algorithm that finds changes in visual code based on the differences of graphical elements.

In 1 study [S33], the authors use model checking, a formal verification technique, to automatically detect game logic bugs.

In 1 study [S34], the authors propose a new Unity tool that can detect bad smells in videogames.

In 1 study [S35], the authors analyse stack trace similarities to identify crash types, enhancing their early detection.

In 1 study [S36], the authors show how imperfect abstraction of the underlying videogame can enhance the performance of the Go-Explore algorithm.

In 1 study [S37], the authors compare different strategies for exploratory testing in videogames. In a follow-up study [S38], the authors present a framework that includes testing strategies tailored for videogames, a recommended order of strategies, and guidelines for applying each one.

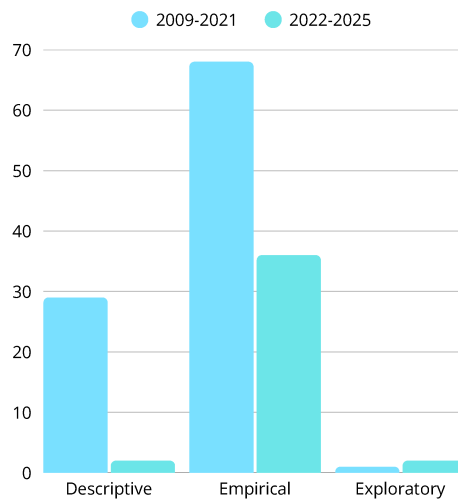


Figura 4.5: Research approaches

4.2.9 Software post-development issues

In 1 study [S39], the authors investigate the potential of visual exploration for debugging and modding using a tool they created.

In 1 study [S40], the authors propose a new technique to reproduce gameplay videos for debugging purposes.

4.3 RQ3: Research approaches

Q RQ₃. *How have research approaches in the domain of industry-scale computer games changed over the years?*

Figure 4.5 shows that empirical studies are overly dominant, as practitioners almost exclusively focus on them. Both exploratory and descriptive studies are almost non-existent for industry-scale games.

4.4 RQ4: Empirical methods

Q RQ₄. *How have empirical research methods in the domain of industry-scale computer games changed over the years?*

Figure 4.6 shows that case studies are still the most used method, followed by experiments. Surveys, on the other hand, are almost non-existent.

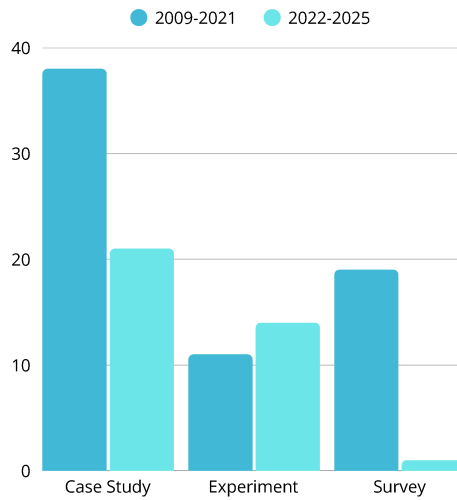


Figura 4.6: Empirical methods

4.5 RQ5: Research quality

Q RQ₅. *What is the quality of industry-scale computer game research currently?*

Regarding the quality questions previously discussed in Section 3.3, Figure 4.7 reveals that every study received a 'Yes' with respect to Q1, the question about motivation. Q2, Q3, and Q4 were almost always answered with a 'Yes', and the few exceptions still get a 'To Some Extent'. Slightly more than half of the studies still have a 'No' in relation to Q5, but this is still a positive trend, as this percentage was much higher in the previous study [1] (slightly more than 80%). There is also a positive trend for Q6, as almost half of the studies received a 'Yes', compared to the previous 20%. These last 2 questions dealt with potential bias and limitations and the generally higher evaluation shows that researchers are starting to acknowledge these aspects of their studies, but there is still room for improvement. Finally, Figure 4.8 shows the distribution of the scores obtained by each study.

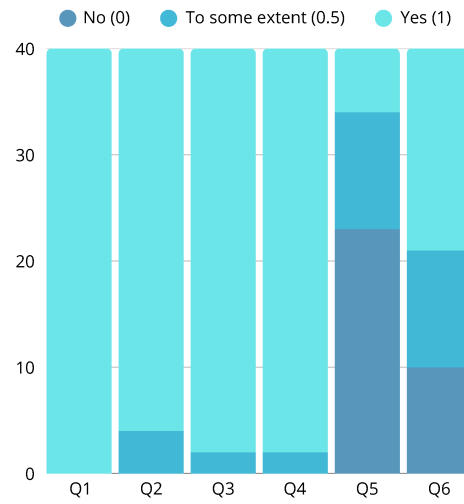


Figura 4.7: Answer distribution

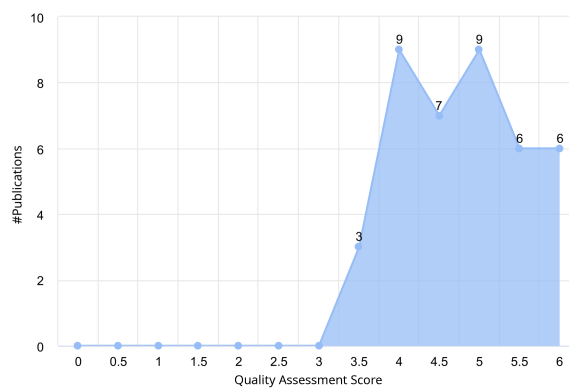


Figura 4.8: Publication quality

Discussion

The next section presents an in-depth analysis of the results of this work. Each research question will be addressed singularly to show details on the most recent trends in academic research. Furthermore, each question will have a dedicated subsection that shows a full overview of the years ranging from 2009 to 2025.

5.1 RQ1: Research intensity

Q RQ₁. *Has the intensity of the research activity on software engineering for industry-scale computer game development changed over the years?*

As seen in Section 4.1, the interest in research in the GSE field continues to grow, along with the revenue of the gaming industry. The distribution of the included articles has changed from what Chueca et al. [1] observed in their review:

- **Venues of publication:** the 40 studies included are spread among 29 venues, 7 of which are dedicated to videogames and amount to 13 papers (or 32.5% of the total). Previously, data showed that 98 studies were spread over 86 venues, with 14 dedicated venues containing 17 articles (or 17.35% of the total). This suggests that the relative weight of the dedicated venues has grown substantially. Dedicated venues such as the IEEE Transactions on Games, the International

Workshop on Games and Software Engineering (GAS), the IEEE Gaming, Entertainment, and Media Conference (GEM), and the IEEE Conference on Games (CoG) appear to be consolidating their role as reference points for GSE research.

- **Publishers:** the total number of unique publishers decreased from 13 to 8, but the studies are still unevenly distributed. IEEE still holds the largest share (37.5%, previously 40%), but ACM closely follows in second place with 30% of the total (previously 23%). Both Springer and Elsevier maintained similar percentages compared to the previous study and are currently around 10% each. Lastly, the number of smaller venues with only one or two studies has reduced from 9 to 4, indicating a gradual centralisation of research outputs in more established publication channels.
- **Types of publication:** conferences are still the most frequent type of publication (from 62% to 60% of the total), followed by journals (from 29% to 32.5% of the total), and lastly workshops (from 8% to 7.5% of the total).

These data suggest that the research community is less dispersed than before and is slowly coming together. The dedicated avenues are starting to get more attention from researchers, at least regarding the industrial scale analysed in this work, but there is still a long way to go.

The increasing academic interest in the GSE field highlights the recognition of videogames as complex software systems and the significance of this field. At the same time, the persistence of a relatively high number of venues indicates that the field is still in a consolidation phase.

5.1.1 RQ1 Overview

As seen in Figure 5.1, looking at the data for the entire 2009-2025 time frame clearly shows how interest in GSE research increased over the years. This trend closely mirrors the growth of the videogame industry, whose increasing economic and cultural relevance has contributed to the perception of videogames as a legitimate and impactful domain for SE research. This alignment between academic attention and industrial expansion reinforces the idea that GSE is a separate domain from SE.



Figura 5.1: Increase in GSE activity

The fact that conferences have always been the most popular publication type can be justified by the focus on an industry-scale level, since they are more prestigious and undergo a stricter peer-review process than workshops. Conferences also offer faster publication cycles, which is an important aspect in the computer science field, where both technological advancements and industrial practices evolve rapidly. Journals remain important venues for more mature and comprehensive studies, but their slower publication process makes them less suitable for reporting findings.

IEEE emerges as the publisher of choice, followed by ACM, Springer, and Elsevier. Their prestige attracts more researchers, but it is also worth noting that their dedicated venues have only recently gained more attention. This suggests that the field of GSE is still searching for its identity, but its growing presence indicates its increasing institutional recognition, which, in turn, may foster more stable research lines, specialised conferences, and dedicated tracks.

As a final observation, the steady increase in research output, combined with the gradual concentration of venues and publishers, suggests that the GSE field is in a phase of thematic and methodological consolidation.

5.2 RQ2: Research topics

Q RQ₂. *How have software engineering research topics in the domain of industry-scale computer games changed over the years?*

The main topic is 5.3.4 *Software verification and validation*, just as before, but its volume doubled in the last few years, jumping from 26.26% to 52.50%. Although most of these papers strictly relate to testing (only suitable category for them), there are still papers that would fit elsewhere according to the previous classification system, contributing to a more even distribution. The second most common topic is 5.1.2 *Software system structures*, which went from 14 to 7 total papers, but maintained a similar percentage, from 14.14% to 17.50%. Both 5.3.1 *Designing software* and 5.3.2 *Software development process management* lost approximately 10% each compared to the previous study, which means that practitioners lost interest in these topics. All other topics maintained similar percentages, showing that they are either not that interesting to researchers or not that relevant to the GSE domain.

This research showed that the generic topics most chosen are, from first to third, *Analysis*, *Implementation*, and *Quality*, while previously the leaderboard was held by *Design*, *Quality*, and *Management*. It is worth noting that Chueca et al. [1] decided to divide their time frame into 6 total sets of 3 years each (except the last comprising 4 years); from 2018 to 2021, *Design* was the first, *Quality* was the second, and the third place was shared between *Management* and *Implementation*. The newly interest in *Analysis* highlights a tendency in practitioners to prioritise works that deal with new methods and metrics. This shift could also reflect the increasing complexity of modern game engines and pipelines, which require more sophisticated tools.

The lack of studies in the *General* topic was expected and suggests that researchers prefer to focus on specific things rather than the broader aspects of GSE, at least at the industry-scale level. Both *Design* and *Management* now count 1 study each, showing that those fields may have been already thoroughly researched in the last years or that their challenges are perceived as well understood.

Implementation is also a rising field, as it counts 12 papers just like the previous work, but condensed in a timespan 3 times smaller. This means that there is a growing interest in producing new tools and frameworks to facilitate development processes, and it is plausible that many more studies will be published in the next few years.

Lastly, *Quality* is also seeing a positive trend, with 10 total papers out of 40 included. This, along with *Implementation*, shows that the focus for the future will be on the aspects that enhance software performance and reliability. This strong emphasis also

indicates that researchers are prioritising work that can directly improve production efficiency and reduce costs.

5.2.1 RQ2 Overview

Looking at the data for the chosen topics from 2009 to 2025 clearly indicates that the 2012 classification system is inadequate for GSE: while *5.3.4 Software verification and validation* takes up most of the published papers, 8 topics count between 1 and 5 papers and 6 topics count no papers at all. Among the other 5 topics remaining, 3 maintained similar percentages, and 2 lost relevance. Since the time frame considered is wide and encompasses many publications, this aggregation of data strongly indicates that the current classification system should be modified to tailor the needs of the GSE domain or that a new one should be made from scratch.

As previously stated, papers focused on testing are placed in the *5.3.4 Software verification and validation* topic, and in recent years almost all of the papers that are in this category strictly relate to that. This clearly shows a great interest in making testing more efficient, fast, and accurate, as it takes a lot of effort and time for development teams. However, since videogame testing refers both to finding bugs and analysing user experience, it could be beneficial to create two different topic categories to address this double nature. Such a distinction would allow researchers to have a better representation of their work.

In general, looking at the data for each generic topic from 2009 to 2025 shows how the GSE field is reaching its maturity. *Design* had always been the most popular topic, except for the 2015-2017 time frame, where it got surpassed by a sharply rising *Quality. Management* was also steadily growing from 2009 to 2021, but both this and *Design* were not addressed by researchers in the last few years. After almost tripling its volume from 2012-14 to 2015-17, *Quality* maintained its popularity over the years. *Implementation* and *Analysis* have only recently gained attention from practitioners, who produced, respectively, 50% and 60% of the total body of papers in the 2022-25 time frame. These trends indicate that contemporary research is increasingly orientated toward practicality, reflecting a field that is maturing and aligning more closely with the needs of modern videogame production.

5.3 RQ3: Research approaches

Q RQ₃. *How have research approaches in the domain of industry-scale computer games changed over the years?*

Exploratory studies are usually present in younger research areas, and this paper shows that they are almost non-existent in recent years. This is in line with what Chueca et al. [1] previously observed and highlights the maturity of the GSE field. The decline in exploratory studies indicates that the founding questions, such as those aimed at mapping the domain, have largely been addressed in earlier phases of the field’s development (as seen in [3]).

Descriptive works also saw a decrease in popularity, counting 2 papers out of 40 like exploratory ones. An explanation could be that researchers have moved closer to industry demands, thus prioritising practical matters. Another plausible explanation is that descriptive studies, while valuable for establishing taxonomies or documenting practices, tend to have limited impact on industrial workflows unless they are paired with actionable insights. This strong focus on empirical methods also reflects a tighter alignment between academia and industry, where evidence-based evaluations and case studies are perceived to be more directly beneficial to improving tools, pipelines, and processes.

5.3.1 RQ3 Overview

Since 2009, practitioners have increasingly focused on empirical research and have progressively abandoned exploratory studies. This is a common trend in fields that move towards maturity, since exploratory studies are meant to shed light on aspects not yet well understood. Now that the field of GSE is more stable, researchers are focused on refining existing knowledge and assessing the effectiveness of specific tools or techniques.

The gap between industry and academia is getting smaller by the year and this claim is also supported by the lack of descriptive works, which are useful from a research standpoint, but generally do not produce anything tangible. Empirical studies, on the other hand, often involve industrial case studies or controlled evaluations that

can directly inform practice. This trend suggests that GSE research is increasingly orientated towards solving concrete problems faced by developers.

5.4 RQ4: Empirical methods

Q RQ4. *How have empirical research methods in the domain of industry-scale computer games changed over the years?*

Surveys kept the declining trend already discovered in the previous work [1] and are almost non-existent in this dataset (only 1 occurrence out of 36 empirical studies). This confirms a broader methodological shift, as surveys, once common in the early phases of GSE research, now appear less suitable to the technical nature of the field.

Case studies are still dominant and amount to almost 60% of empirical studies, suggesting that industry collaboration remains central to GSE research. In-depth case studies strengthen the bond between industry and academia and provide data and context that help to reduce the gap between modern videogame development and theoretical advancements.

Experiments, on the other hand, are on the rise, taking for themselves almost all the space lost by surveys. This trend was already expected to increase since 2009 [3], but that did not happen: a possible explanation was that, since the game industry is fast-paced, a controlled experiment required too much time and effort. However, in recent years, experiments have started to gain attention as a way to evaluate new tools and methods.

5.4.1 RQ4 Overview

Looking at the entire data from 2009 to 2025, we can see the effects of the closing alignment between industry and academia. The fast-paced and practical videogame industry demands actionable insights, which means that the broad, high-level information produced by surveys is less valuable than those produced by case studies and experiments. Another possible reason for the loss of relevance of surveys could be the availability of alternative data sources, such as telemetry and user reviews on platforms like Steam.

Case studies became the most popular research method and will most likely continue to be so in the future. Their ability to capture real-world complexities makes them indispensable for understanding how videogame development practices evolve and how teams adopt new tools.

However, experiments have recently regained some attention as an alternative for testing and analysing new tools and frameworks in a more thorough way. This may indicate a methodological maturation, as researchers are increasingly focused on measuring effectiveness, comparing approaches, and establishing empirical baselines. This trend may lead to stronger evidence and better reproducibility.

A final observation concerns the coexistence of case studies and experiments, as their complementarity can make the GSE field more balanced and give more variety to research. In the future, they are expected to maintain the same relevance, as they are both practical approaches that provide data and context that help reduce the gap between modern videogame development and theoretical advancements.

5.5 RQ5: Research quality

Q RQ5. *What is the quality of industry-scale computer game research currently?*

The maturity of the GSE field is reflected by the higher quality of the studies included in this review: their average score increased from 3.49 to 4.8. As mentioned in Section 3.3, this score is related to the quality of the reporting of the findings rather than to the intrinsic quality of the research itself. However, the improvement is substantial and indicates that the authors are increasingly adopting clearer structures and becoming more rigorous and transparent.

The higher score is mainly due to a better explanation of the limitations of the studies, which became clearer and more organised in recent years due to the increased use of standardised methodologies. In more than half of the studies, the authors are still not addressing their own biases, but a positive trend can still be seen, as the average score for that particular question (Q5) went from 0.15 to 0.29, almost doubling. Although the value remains low, the improvement is notable and indicates

that some researchers are starting to recognise the role of their bias in shaping the study design.

5.5.1 RQ5 Overview

Since 2009, the quality of the reports in GSE has continued to increase, showing that researchers have started to be more clear in their explanations and adopt best practices. This improvement aligns with the broader evolution of empirical software engineering, where structured reporting guidelines have become widely adopted.

The increased consistency in the way studies describe their context, participants, and data sources suggests that GSE is approaching the reporting quality seen in more established areas of Software Engineering. Such convergence is essential for enabling cross-study comparisons, meta-analyses, and the accumulation of reliable evidence.

Another reason for improvement in reporting quality may be the growing involvement of industry partners, since these collaborations often require clearer documentation and more structured communication. As a result, academic researchers might adopt higher reporting standards to meet these expectations.

Finally, this upward trend suggests that GSE is moving towards a more standardised research culture, which will likely support future advancements in the field, enabling more reliable comparisons and fostering cumulative knowledge building.

Key insights

RQ1: Research activity increased, with 40 publications in less than 4 years. The community is coming together, but there is still much room for improvement.

RQ2: GSE is evolving through close contact with the gaming industry, veering towards practical matters, so topic interest is shifting accordingly.

RQ3: Empirical studies are by far the most common type of research and greatly aid in bridging the gap between industry and academia.

RQ4: Case studies are still the most popular research method, but experiments are on the rise, as they can also help provide practical data.

RQ5: The quality of the reporting of the findings has increased greatly, but researchers can still make improvements to address their biases.

Future directions

In this section, Chueca et al. [1] presented challenges and open problems in the field of GSE. This paper intends to expand this front, analysing each subsection to show if and in what measure they have been addressed.

6.1 Cross-fertilization between GSE and MDE

Game engines are an essential aid for videogame development, as they can both ease the work of developers and allow people with no programming knowledge to contribute to the creation process. Model-Driven Engineering (MDE) is a methodology that focuses on domain models, conceptual models of the topics related to a specific problem, to implement a system. Its core principles are abstraction, simplification of the design process, and the promotion of collaboration between different individuals and teams. The MDE community could find perfect examples and case studies for their research in game engines, yet practitioners do not consider them much. Recently, 3 studies that analyse the impact of Model-Driven Development (MDD) on the development of videogames were published [S7, S10, S15].

6.2 Game software product lines

Software Product Lines (SPLs) are a set of methods, techniques, and tools used to create a collection of software systems that share core assets. They enable massive and strategic reuse of components to improve quality and speed of development. Software reuse is also an important aspect for videogame development, as it can be helpful in meeting deadlines. However, this urgency hinders the investment of time in developing specific techniques for code reuse. In recent years, a study investigating general reuse practices has been published [S17], while 2 studies focused on the benefits of using SPLs [S15, S16].

6.3 The blossoming of publishing platforms

In the last decades, it became increasingly normal to have videogames published on multiple platforms, either since day one or with a later porting. This means that developers must, at one point or another, adapt their code to fit different input schemes and hardware capabilities, while maintaining high quality. This task is non trivial and requires a lot of time and resources, but the use of game engines can make the process easier, leaving out just the adjustments for performance. GSE research did not make any progress on this topic in recent years, so there is ample room to develop supporting tools that help developers port videogames across platforms.

6.4 The multifaceted nature of GSE

A traditional problem in software engineering is the generalisation of findings, and this is exacerbated in the GSE field, given its heterogeneity. There are many factors that must be considered: the size of the team behind the videogame, the platform where it was published, the engine used, the genre, etc. Recent studies did not attempt to bridge this gap, so this is a rich opportunity space.

6.5 Planning and scheduling issues

Videogame development teams generally employ agile methodologies when working on their products, given how fast-paced the industry is. The tight deadlines for publishing a videogame often cause teams to enter crunch time, periods in which people work overtime to meet the expected release date. This is such a common occurrence that the term crunch culture is often used when discussing videogame development settings. There is a strong need for better adaptations of agile methodologies, both for the mental and physical health of programmers and for the quality of the products. However, in recent years, no studies on this topic have been published.

6.6 The consequences of games as a service

In recent decades, the business model called Games as a Service (GaaS) became increasingly popular [22]. Similarly to the concept of Software as a Service (SaaS), these videogames are treated as evolving products that will be maintained for a long time. Leaving aside the various strategies to monetise these videogames, GSE practitioners should focus on maintenance and management aspects, since GaaS are expected to at least stay as relevant as today for the coming years. However, as seen in Section 4.2, these aspects have not been of interest in recent years, leaving much room for research.

6.7 GSE security

A large portion of the user base of videogames plays Massive Multiplayer Online games (MMOs). These videogames must have a robust, secure, and reliable network system that can handle large volumes of players. The amount of calculations and interaction that need to be tracked can cause latency issues, which is even more concerning for e-sports tournaments, where professional players compete to win prizes. From an architectural standpoint, 2 studies on improving Quality of Experience (QoE) for players were recently published [S2, S3]. Another concern for MMOs is cheating (exploiting bugs or using various tools or methods to gain unfair advantage

over others), since it negatively impacts the experience for other players involved. An exploratory study was recently published to investigate current awareness and priorities in regard to security in videogame development [S9].

6.8 Weaving software and art design

One of the challenges of videogame development is documenting phases and processes, due to the multiple disciplines involved and the tight schedules. A common practice involves writing Game Design Documents (GDDs), and many papers on how to do so have already been published. However, there is currently no established standard and, since they have a static nature, practitioners may focus on new, more flexible, options.

6.9 A new wave of requirements research for GSE

Among the requirements for videogames, one that stands out is related to fun and entertainment, since traditional SE does not bother with this. Ampatzoglou and Stamelos [3] found that Requirements Engineering (RE) was the topic most studied before 2009, but then Chueca et al. [1] found that between 2009 and 2021 this topic started to lose appeal. In recent years, no new studies on RE were conducted, but it must be noted that during the selection phase, various articles dealing with user experience were found, but their main contribution was to Human-Computer Interaction (HCI).

6.10 Testing and debugging in the face of gameplay subjectivity

Closely linked to RE, testing in videogames is peculiar because of each player's idea of fun. This means that testing can be divided into two main groups: the first one searches for bugs and errors in the program, while the second, gameplay testing, focuses on the user perception. Regarding the latter, a study on determining the

comprehensive difficulty of enemies has recently been published [S8]. Regarding the former, this was the most discussed topic in recent years and will probably maintain this intensity in the coming years, as guaranteeing high quality products is one of the main concerns of videogames development.

6.11 Green GSE

One of the most important factors in a videogame is performance: having efficient code means providing a better and more stable experience for users. Adopting green methodologies can reduce the carbon footprint of the gaming industry, positively impacting both developers and users. Understanding which components of a videogame consume more energy means understanding which aspects could benefit more from green practices. While writing fast code is an open issue, knowing critical areas is already a step in the right direction. This field is starting to attract researchers, as 2 studies dealing with energy consumption were published recently [S1, S10].

6.12 Search-Based GSE

Search-Based Software Engineering (SBSE) applies metaheuristic techniques such as genetic algorithms to software engineering problems. A way to apply this approach to GSE is by leveraging Non-Playable Characters (NPCs), since they can be used to perform simulations and improve the fitness function. Since the user base of videogames is huge, players can contribute to the vision of Humans as a Fitness Function [23], where humans help evaluate solutions found in SBSE. Recently, a promising study on Traceability Link Recovery (TLR) in GSE was published [S29].

6.13 Content creation and artificial intelligence

Content creation and Artificial intelligence were out of the scope of this review. A lot of research is being conducted in these two domains, which means that there is plenty of material to analyse to determine the state of the art and how present GSE is in them.

Threats to validity

Wohlin et al. propose a classification to assess the validity of the results with four categories of validity concerns [24, 25]. This section is meant to test the study against them.

Internal validity: this category deals with factors that affect dependent variables without the knowledge of researchers. Meticulously defining the criteria used for the inclusion of a paper and its subsequent evaluation should mitigate this threat.

External validity: this category deals with the generalisation of the results. To collect enough studies to be representative of the population, two databases, IEEE and ACM, were used in conjunction with an indexing database, Scopus. Another aspect to consider is that only industry-scale computer games were considered, so it may not be safe to generalise results to the whole field.

Conclusion validity: this category deals with the relationship between the theory and the outcome of an experiment. To mitigate this threat, best practices were followed [18, 26, 27]. However, it must be noted that this work was not done by a team but by a single researcher, which means that some bias could still be present.

Construction validity: this category deals with the relationship between concepts, theory, and measures that researchers have in mind. To mitigate this threat, both the research questions and the search process were thoroughly explained.

CAPITOLO 8

Conclusion

This work presents an updated SLR to study software engineering research for industry-scale computer games. It includes 40 studies that cover the years from 2022 to almost all 2025 and shows the current relationship between industry and academia.

The findings show that interest in GSE continues to grow and that the field is steadily maturing. Compared to previous results, the overall quality of the studies has improved, specifically with respect to methodological clarity and reporting standards. However, there is still plenty of room for improvement, especially regarding the critical examination of results and the articulation of threats to validity.

A key insight concerns the limitations of the 2012 ACM CCS, which does not correctly represent the GSE research domain. The data reveal that in recent years 11 out of 20 subcategories contain no studies at all, while the 5.3.4 *Software verification and validation* topic includes 21 out of the 40 studies. As stated previously, since this is the only topic in which papers related to testing can be placed, it could be beneficial to create two new specific topics to account for software testing and gameplay testing. The fact that the remaining 19 papers are distributed in 8 categories indicates that there is at least some diversity in the topics.

This distribution also proves wrong some of the predictions made in the original

study by Chueca et al. [1], who expected architecture and design techniques to be the topics most investigated. This contrast also supports the claim that GSE and traditional SE are diverging, which opens the possibility of a new classification system tailored for GSE.

The research community is still somewhat dispersed, but the increasing visibility of specialised journals and conferences indicates a trend toward consolidation. These venues are expected to become more and more relevant in the coming years.

The prevalence of empirical studies suggests that the gap between industry practice and scientific research is narrowing. This claim is also supported by the fact that case studies are the most chosen approach, followed by experiments. Many studies are grounded in real development environments that often involve collaboration with videogame studios. This trend strengthens the validity of the findings and shows the close connection between industry and research.

The widespread availability of game engines and the popularity of some of them, such as Unity and Unreal, have lowered the entry barrier to videogame development. This allows small teams to accumulate practical experience quickly, but this accessibility does not guarantee familiarity with software engineering practices. Such gaps present an opportunity for engines to integrate and promote good SE practices directly within their tooling and documentation.

Research can also help smaller studios to work better and more efficiently by improving automation and making testing practices more accessible. This would allow developers to allocate more time and resources to other aspects of the development process, such as the creative side of it.

Ultimately, this study has two goals. The first is to provide fresher background information for any relevant future work, helping the community explore the research gaps that GSE presents. The second is to create a bridge between industry and academia, as they can positively influence each other: industrial practitioners can share their expertise, while researchers can find interesting cases for their studies.

As videogames continue to push technical and creative boundaries, Game Software Engineering remains a fertile ground for advancing both software engineering theory and practice.

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